

ANUPREET SINGH

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EDUCATION

University of Maryland, Baltimore County

(Master of Science in Computer Science, GPA: 3.8/4.0)

Baltimore, MD

Aug 2024 – May 2026

Indian Institute of Information Technology, Sonapat

(Bachelor of Technology in Computer Science and Engineering, GPA: 8.45/10)

Haryana, India

Aug 2019 – May 2023

SKILLS / RELEVANT COURSEWORK

Programming Languages: Python, C++, Go, JavaScript

Tools and Frameworks: Docker, Kubernetes, GitHub Actions(CI/CD), Git, AWS, Azure, Terraform, Linux, Redis, Kafka, PyTorch, LangChain, Pinecone, PostgreSQL

Areas of Interest and Coursework: Site Reliability Engineering, AI & LLMs, Distributed System Design

INDUSTRY EXPERIENCE

Data AI Engineer | Platinum Business Services (R&D Team) | Catonsville, MD

Dec 2025 – Present

- Containerized the platform (FastAPI, Postgres, Qdrant, Ollama) and deployed it to **AWS EKS** with Kubernetes manifests and a GitHub Actions **CI/CD pipeline**, adding health checks and admin endpoints to download models and rebuild search indexes, so the team could manage models and indexes without shelling into pods.
- Diagnosed out-of-memory failures in the Ollama **container**, tracing it to freshly pulled 4-5 GB model weights lingering in the OS page cache, and fixed it with targeted **posix_fadvise** eviction through a shared volume, **eliminating model-load failures**.
- Developed backend services for an **AI-native** GRC platform that audits an SSP draft for compliance with NIST 800-53, chunking ~1,200 controls into a per-model vector index with linkage metadata tying each chunk to its parent control and sibling clauses, cutting a control-by-control review from **days of manual work to <10 minutes**.

Software Engineer (AI) Intern | PMA | Tallahassee, FL

Sep 2025 – Dec 2025

- Architected a fully serverless review pipeline on **AWS** (S3 + CloudFront-hosted React SPA, Django on Lambda, DynamoDB, and an SQS-triggered Lambda AI worker) that retries failed analyses off a dead-letter queue and logs per-report model API spend, sustaining 200+ submissions across 5 cohorts at **~\$3/month** idle cost.
- Built an **AI-powered** PRD review platform in React and Django that compares each uploaded draft PDF against a reference PRD and emits schema-validated gap flags for a human-in-the-loop reviewer to confirm or dismiss, giving a cohort-wide matrix that replaces a read of every draft and cuts feedback-session prep from **8 to 2 hours**.

Software Development Engineer | Turfco Landscape Co. | Surrey, BC

Jan 2023 – June 2024

- Designed and deployed the **AWS architecture** that took the business off a WordPress brochure site, with SPA on CloudFront, autoscaling Express tier behind an ALB across two AZs, Multi-AZ RDS, sized for ~10× seasonal peaks and load-tested to **~50 quote requests/sec at <120 ms p99** with zero-downtime rolling deploys.
- Architected a self-serve quoting platform that priced jobs with deterministic multi-stage evaluation, JSONB encoded conditions, per-quote audit trails of every rule applied, and a write-invalidated **Redis** cache, all driven by rules editable from an admin console instead of a developer deploy, cutting quote turnaround from **~2 days to <1 min**.
- Built the online ordering and booking platform by taking self-serve orders through Stripe Checkout deposits, confirming them with signature-verified idempotent webhooks, and relaying a transactional outbox to Kafka workers for order confirmation, customer notification, and crew allocation, taking **\$600K+** in online bookings across ~1,100 orders.

PROJECTS

Role Radar | macOS job-alert app with serverless AWS failover

- Built an async **Python job-ingestion** engine that tracks 7,700+ employers' careers pages, auto-selecting 1 of 22 source-specific scrapers (Greenhouse, Workday, etc) by URL and routing every request through a robots.txt-compliant client with per-host rate limits and Retry-After backoff, checking ~3,600 companies per 18-minute pass across ~1,000 hosts at a **0.18%** request failure rate.
- Designed a **laptop-to-Lambda failover** that keeps checks running while the Mac sleeps, electing a single runner through a conditionally written DynamoDB lease whose epoch acts as a fencing token on each company's transactional save, so a laptop waking mid-run can never overwrite its successor's state, with Lambda taking over within **8 min** and handing back in **~15 s**.